AI & Cloud Infrastructure Space – Astera Labs, Broadcom & Nvidia – Former Senior Executive at Dell Technologies Inc

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Specialist Background

- Extensive experience in the technology sector, focusing on AI, cloud infrastructure, communications and IT
- Having worked at Dell, is well-acquainted with its VDI and DaaS pricing model, product offerings and revenue drivers
- Well-placed to discuss companies that produce semiconductors and related technology, especially Marvell, Astera Labs, Broadcom and Nvidia

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Transcription begins

Analyst:
Astera Labs has got three product families, the retimers, the Smart Cable Modules for the active electrical cables and then the CXL modules. Jensen had plenty of comments coming out of GTC, talking about the move towards an increased use of copper as it relates to its NVLink interconnects, and so there have been a lot of questions as we move to the latest generation of Nvidia's GPUs, particularly those GB200s.

It seems there's some misplaced concern that Astera Labs is going to lose retimer content, but my understanding was that NVLink interconnects didn't require retimers in the first place, and so it's not like they're getting designed out of those types of architecture deployments and that, on the front end of the motherboard, there continues to be a need for those retimers and with the shift up in PCIe Gen speeds, that there might even be an increased content opportunity. Can you help level-set? Is there anything that I'm misinterpreting as it relates to the introduction of more copper, but maybe it's mostly back end and so not overly impactful to Astera?

Specialist:
I think that's probably their primary headwind. The way you're thinking about it is accurate. There are the B100, the B200 and then the GB200. The GB200 is the one that has the NVL72. It's going to be a Grace Blackwell architecture, and it has that silicon interposer. The fact that they're going to pack such high density into the rack, they're going to avoid optics and retimers to save power. First of all, the GB200 is the last one of that Blackwell series to come out, and it's my understanding anyway that it won't come out until H2 of next year. The Intel or AMD front end systems with the B200 are going to come out, I think, late Q1, and the B100s will be available in early Q1, late Q4, that kind of time frame. First of all, the market is going to get fairly saturated with B100s and B200s before the GB200 is available and in that form factor. It's all about rack-level density for GPUs for foundation model training. Definitely, some sort of impact to the market, but in that same time frame, you got to think about the rise of inferencing and people putting a lot of this generative AI stuff to work. In that case, retimers are still going to play a pretty big role in the scale-out networks, etc. While there will be some impact, I don't think it's going to be as large as some people might think it is.

Analyst:
You mentioned that retimers are continuing to play a role in inferencing. Can we then assume that you wouldn't really expect those NBL72-type deployments to be for inferencing use cases, that it's more for the foundation model training at the end of the day?

Specialist:
Yes, absolutely. I don't think you're going to see a lot of 72-plus GPUs in a rack in the inferencing world. I think that those systems will still have high scale, but I think a lot of them are going to be Intel front end because they're going to be doing multiple tasks. They're going to be guard-rails and RAG and a

whole bunch of other things running there. I think you'll still see healthy demand for retimers in that space.

Analyst:
In terms of the IPO journey, Astera's management team were pretty adamant about talking about not having much of a presence in the back end interconnect portion of the GPU box deployments whereby if you're talking about the NVL switch chip interconnect aspect, there wasn't a retimer there in the past. Is that true, or is management maybe mischaracterising that?

Specialist:
I think the retimers were in the back end. You've got the scale-up, which is the inside cluster networking. To network all the GPUs inside the system, you've got to scale out where you're tying multiple systems together or even clusters. Then you have the front end, which is the regular Ethernet. Definitely, they were in the back end for the scale-out where things were extended because a lot of that was fabric-based. If you look at these servers, I don't know exactly the dimensions, you can look them up, but they're 1.5 metres long. They're pretty long. It looked like mini coffin. The extension of them would require retimers.

Analyst:
In terms of an old HGX box, roughly, what were the number of retimers and how many of them might have been on the front end of the motherboard vs how many of them were part of the back end scale-out aspect, just to understand what that foregone opportunity might end up looking like, based on those NVL72 deployments?

Specialist:
It depends on the specific server, and they're different for different servers. They come in 8-, 16-lane packages. Each one of those GPUs, I think, required eight lanes.

Analyst:
Broadcom is now looking at the space a lot more closely than it had in the past. Marvell is talking about it a little, but maybe not as a big, serious contender. The IPO journey was like, "We've got this 90%-plus share of overall retimers out in the server market, and we would expect to keep 70-80% share over the longer term." That seems to be contradictory with someone like Hock Tan at Broadcom who doesn't really enter markets, if he's going to be the number two guy. It's still early days, we know that might have only just started to be sampling, and we don't even know if that's something you've started to think about because it's not really required to think about replacing the retimer. Is it fair to think that share erosion is probably going to fall somewhere in between there, maybe where Broadcom is not going to absolutely dominate them, but Astera is going to lose some share?

Specialist:
Yes, I think you're thinking about that correctly. Broadcom is aggressively pursuing this. They really didn't pursue it before. First of all, they were a customer of Astera. I think you could assume that that's going to fade away as their product matures. They'll start including their InfiniBand and their switches, their own retimers. That's one area that's certainly going to impact their business. Then Broadcom is very aggressive. You see that across all their businesses, and they know this is something that's important, and so they're going to attack it pretty heavily. Of the threats, I think that's probably a larger threat, and it really is Broadcom. Marvell, other folks have stuff, but they've been out a while. For whatever reason, Astera has been able to dominate the market. I think that you will see Broadcom aggressively pursue the market. The things that are going for Astera there are that it takes a while to qualify the capabilities. In terms of technical qualification testing, Astera, they've got a lot of experience here, so they've got the power of incumbency. With their IPO, they'll have more capital available to be

able to effectively compete and fight back against aggressive Broadcom adoption. I don't know how much the market Broadcom will be able to take. I don't ever want to underestimate those guys. They're a pretty driven company, and when they want to do something, and I think they do want to do this, they're going to be very, very aggressive.

Analyst:
It seems that there are instances where if you're going through the DGX Cloud partnership and you're just getting everything that Nvidia is offering co-signed, it's not really customised per se. Then there are other instances where you would go through the ODM like a Foxconn and do a lot of customisation. I would assume more of the overall GPU demand is probably going via ODM, and so there's plenty of customisation opportunities. That should play in Broadcom's favour because Astera would say that, "We're closely aligned with Nvidia, and reference designs will prefer us." There is a bit of scepticism around how much that could actually insulate them, given that the decision is at the ODM level at the end of the day. What are your thoughts?

Specialist:
Yes, it really is at the ODM level. Nvidia will have a lot to say about that. Again, Broadcom will be aggressive. Now, one of the things to think about is Nvidia and Broadcom's relationship. There's a lot of human. Nvidia and Broadcom somewhat compete with NVLink on a NIC basis. Nvidia, I would say, would be probably incented to continue to use Astera where they're using them.

Analyst:
In terms of the custom ASICs that the hyperscalers are working on, it doesn't really seem like an edge play per se but more for their internal workloads. Is it fair to assume that maybe that's an area that Broadcom could probably be pretty aggressive trying to take share from as well?

Specialist:
Yes. They could be pretty aggressive in combining and incorporating the retimers into their NICs and switches directly, but they got to get qualified. I don't work for a hyperscaler, but I imagine their processes are like ours, and so that qualification process takes time. Astera has the power of incumbency. They're limited. That's a niche kind of thing in the retimer thing. It's not like Microsoft where they can bring in a lot of commercial leverage, but they do have a lot of technical experience and could stay ahead. Broadcom is making big claims they're using a more advanced semiconductor process, etc. They're making some pretty big claims on performance. We'll see. With these products, to me, it feels like you meet the spec or you don't meet the spec. It really revolves around reliability and making sure that you can deliver on what you want.

Analyst:
Broadcom is going on 5nm. Astera is getting on to more leading-edge nodes on their roadmap, but when they announced that Gen6 retimer, they have not come out and said what kind of node it's going to be on. Regardless of that, it was interesting to see that Broadcom talked about 13W power dissipation vs Astera was 11W. It seems even though Broadcom maybe at a more leading-edge node, Astera is somehow still beating it on power dissipation right now. What are your thouhgts?

Specialist:
I would think that it's pretty close. The power is very, very important, and the training set, probably a little less important on inferencing. I think Astera will be able to keep up with Broadcom. Again, they've got experience and the power of incumbency in terms of performance. I think the bigger issue is Broadcom using leverage and cross-leveraging this in the products and how much business Astera does directly with Broadcom which they will lose. I think they'll be able to maintain their market share with Nvidia, which, I think, is primarily the biggest issue.

Analyst:
You mentioned you can cross-leverage, and for Broadcom, that would be bundling it, so to speak, so that, optically, it looks way cheaper than the USD 40 or USD 50 that Astera is offering. Do you see that having an impact on how Astera would have to price its retimers, given that it doesn't have as much of a bundling opportunity, at least at the moment?

Specialist:
As long as the demand remains where it's at, I don't think there's going to be too much of a pricing competition. I think it's going to be based on technology. Training is all about performance. Now, as we get into the larger phase of this, Gartner predicts that 80% of the generative AI boom is going to be in inferencing, and I think they're right ultimately or probably can be higher than that. As we move further into this space, I think that price competition across all of this may enter in for two reasons. One, supply will catch up to demand ultimately, and then the customers are going to change. They're going to move from folks building these models and training into folks using the models, and so productivity gains and costs start to matter a lot more in that space. I think that that will start to appear, but it's still 2-3 years away before we get to that level of maturity in this market.

Analyst:
I assume the bluebird opportunity would be playing a bigger role in the inferencing fabric. It seems the issue is remaining as it's nowhere near where it needs to be on the latency front to actually be efficient. How is your thinking around CXL as potentially the new memory fabric evolved for you, as we've seen some developments play out? I assumed, around two summers ago, that none of the CXL stuff matters until we get these switches that probably don't happen until 2027, and I'm curious if that still remains the case. In turn, this Leo product that Astera talks about is maybe irrelevant for a few more years still. What are your thoughts?

Specialist:
I believe folks are going to start to launch CXL memory fabrics because if you combine DDR with CXL memory, you can actually get better overall performance. Part of this ties back into the inferencing thing. I know I sound like a broken record on inferencing. Inferencing is much more memory-bound. There are other workloads, I think, that are coming that are memory-bound, that are coming out of edge and other spaces. If you interleave DDR with CXL memory, you can actually get superior performance. In the fairly near future, still in the late Gen5, in the beginning of Gen6, in the Gen5 time frame and coming in Gen6, on PCIe, you're going to see launches of CXL-based memory. Today, CXL, I think the hyperscalers use it primarily to share memory across servers and solve the stranded memory problem. I think that you'll see a lot more of that happening because edge requires that as well. Not as near-term, but as edge grows and gen AI is going to accelerate that growth to get to the modern multi-workload, multimodal edge, it will benefit from CXL as well. Actually, I'm a CXL fan. I think it's got a good future. PCIe, in general, will increase, and there will be a Gen7, etc. It hasn't realised it's prominence, but even ultra Ethernet and some of these other more exotic networking technology being developed around AI today are going to require retimers, and memory fabrics, I think, are definitely on the horizon and need to be on the horizon.

Analyst:
In the sense that these workloads are memory-stranded, the reason that my baseline was that the actual CXL 3.0 switches are what's really going to help, is that the pooling capability of it taps into way larger capacities of memory at one time. You're talking tens or hundreds of terabits, whereas these CXL controllers that Astera offers is one or two terabits. Maybe you're deploying two to four at a time, so it's closer to 10, making enough of a difference for the memory bottleneck. Can we assume that this will really take off when you can use that pooling element and tap into just orders of magnitude more memory at one time?

Specialist:
I think it will make a difference. These things have cost, and CXL is fairly mature. The complexity of expanding these memory fabrics, again, maturity will count for a lot here. Over time, things may evolve to different types of fabric technology. The ultra Ethernet guys would tell you they're not focused on it. They're focused more on scale-out networks to do intercluster networking and pool that stuff together. Right now, I think it's the game in town, and inferencing and other workloads are going to be memory-bound and CXL can help a lot there. It's going to be implemented in these hybrid nodes. We'll see how well that gets adopted here fairly shortly, actually, I think, into next year.

Analyst:
Who else has decent-quality silicon? Micron and Hynix have talked about CXL controller products. I'm not sure if they're licensing that from Astera or if that's competitive vs what Astera is doing, but thinking about who they're going to compete with to get these Leo controllers adopted, who comes to mind?

Specialist:
The processor guys have all adopted it. They're building the capabilities into the CPUs. Micron is committed to it. I don't know what Samsung is doing. I haven't checked to see if they're going to play in this space as well. Astera could get the majority of that because they've got products out, but the memory guys will play. It's going to be hard to see how that breaks out over time.

Analyst:
That's the next catalyst to think about is how much this stuff really starts to pick up. Astera would tell us that they're collaborating with all the memory suppliers, but I don't see a need for them to partner there. I would just assume that they come out with their own controllers and pursue the market head on. Is there a mutual, beneficial dynamic where Astera would actually be able to partner with these guys and rise the tide together?

Specialist:
I think they would be able to partner with them. They don't really compete head on except in that space. Again, the memory guys all got squeezed really hard post-pandemic and coming out of that, and so they cut back on a lot of programmes because of the fall and the client business dropped dramatically, mainly post-pandemic hangover, stuff like that. That business is always up and down, but, and it's worth researching, I don't know, they may be counting on OEM-ing a lot of that stuff from Astera, either licensing IP or directly OEM-ing it from them. It wouldn't surprise me if at least some of those folks are doing that.

Analyst:
On the actual device side of things, Microchip has been doing DIM controllers for a bit, Marvell acquired Tanzanite and Rambus licenses its IP. I don't really see Rambus as a head-on competitor. It seems like this is maybe a pretty similar situation where Astera is first to market, the only one with solid product and so, by definition, would dominate share of the early adoption curve. What are your thoughts?

Specialist:
I think so. There's doubt hanging over CXL. I think it's going to serve a need. A lot of other folks may be rethinking how they attack CXL and Astera is already there.

Analyst:
How will the market shift once we do get to the CXL switch chips at the end of the day? Broadcom is holding its CXL switch roadmap, Xconn is a private start-up that floundered on its first iterations of

silicon. I believe that'll change the opportunity a bit. Expansion is a little different than pooling. How does that impact Astera's role in the CXL opportunity?

Specialist:
CXL was going to be the everything-everything, and it's certainly not. It's going to be targeted toward memory expansion. They're better solutions to do the extension of devices and GPUs and stuff like that with the proprietary solutions and, ultimately, ultra Ethernet and other non-proprietary solutions. It would be focused on memory where it has been, but that's my view. I think that market will grow. I think that generative AI is going to do nothing but increase the size of that market. I think that'll thin the field, and again, folks who have the IP that are already there should be able to dominate. Astera is one of the early players there. The question is, "How big will that market be? Does somebody like a Micron or SK Hynix or Broadcom or somebody else try to enter it early?" A lot of that dynamics will really be on the timing of inferencing and how distributed the inferencing market is. I think that that is going to be a huge market, myself. I think there's a lot of need to place the, at least, some of the inferencing capabilities near the data source. That stuff is going to be much more memory-bound than the training, which is almost all computationally bound which ties back to networking.

Analyst:
You mentioned that inferencing is going to be 80% of that and a lot of it driven by the edge. What is the split of inferencing instances that would be by the data source? You've got 30 million server CPUs out there, a tonne of them are going to be inferencing five years from now, and 50% of those are going to need help on the memory side of things because they're close to the data source. All of a sudden, you tack a controller or two on those and put an ASP on it, and you can get to a market opportunity roughly. How much of inference will need to be closer to the data source?

Specialist:
It's use case-dependent because there's still search and the social media stuff, which is the dominant use case now for gen AI. If you think about virtual chatbots, synthetic data generation, the use in analysis of corpus of words, things like that, you're going to want a lot of that very near. What I see is a shorting of functions in the models. The tokenisation, the embedding, the RAG and the guard-rails are what you're going to want to put close to the data source, and that's a pretty good percentage of the transformer architecture. The core model would still be data centre-based. Those other functions, the vector databases will have to be distributed in a two-tier model for doing the RAG-based work. I think a pretty high percentage, 60% would be a guess, and that 50-70% range is going to end up being distributed ultimately. Most of these use cases, you're not going to want to have to haul all that data back locally, and you've got another issue that's driving that, this is edge 101, which is latency. A lot of those use cases are going to be latency-bound, particularly the virtual chatbots and all. You're going to need to be able to do a lot of the prompt augmentation locally, not just because of the intensity of the data, but the latency.

There's just the fact that, I think, generative AI caught everybody by surprise, but the hyperscalers are having a little bit more of a challenging time responding to it for a couple of reasons. One, in the training world, these are not shared, just the obvious fact. When it takes six months to train your model, you're going to be running it non-stop, run to completion. That's untypical to their business model of sharing infrastructure with customers and having a statistical gain. The other issue, of course, is that their data centres are not architected to do this kind of density of computation, and it's really hard to retrofit the data centres to make that happen. It's almost easier to build them from the ground up. In any event, given these facts, I think a lot of the inferencing will push out of the data centre to the edge. 60% would be my guess, and I think that's relatively conservative for long term. How big is that? There's a lot of forecasting on edge and sizes. I think all those forecasts are going to go up as a result of this, and the analysts should be resizing those things here shortly, if not already publishing stuff.

Analyst:
It seems like nobody has come up with a good way to talk about what the return on invested capital is for the actual end customers, not the hyperscalers themselves, but the end customers with all this GPU demand.

Specialist:
I think you're absolutely right.

Analyst:
It seems some people are taking the more Draconian side of things. A partner at a venture capital firm said that there's a USD 600bn hole that's not able to get filled, but I don't think that it contemplates the productivity gains and the cost savings where you don't need some of these call centres or customer centres. It saves plenty of companies tens, hundreds of millions of dollars. If you bundle all up Fortune 500 enterprises all using this stuff, you get to several billion dollars just there without even thinking about the revenue opportunities. What are your thoughts?

Specialist:
I think the business cases are very attractive. The previous technical term, you can tell. There'll be a lot of use case dependence on these things, but there are going to be tremendous productivity gains. Everyone is thinking virtual chatbots, I actually think that's one of the harder use cases to do because human interaction is just so complex, and fundamental to the transformer architecture is hallucination. It cannot be solved very easily. We almost need a new architecture. That's a harder use case. Sentiment analysis, looking at large corpuses of words, synthetic data generation, multimodal capabilities, there is a very long list of use cases that this technology is going to be very beneficial for. I think some companies are starting to lean forward and saying, "Let's let's force this transition, and let's start to take the productivity gains as we implement the technology." I think we're going to start to see that. I think there are lots of use cases that are going to appear with these models.

Analyst:
It seems like everyone likes to talk about how the telco CAPEX was back in the early 2000s as a way to think about this GPU demand for several reasons, one of them being the fact that hypers might have to look at greenfield builds and they've already announced a tonne of them having data centres that do this type of computing effectively, and that this buildout definitely seems to have legs into '25, if not beyond that. Is this your sentiment?

Specialist:
I think it does. One thing I would be cautious about is that, right now, we're seeing the training demand, and ultimately, inferencing is going to be much larger than training, building integrated development environments to actually deploying the software. Maybe 1% of the CPU capacity of the world is used to develop, so 99%-plus is used to deploy it and operate it. It'll be a little bit different than that because training is so computationally intensive. Gartner thinks it's 80/20. The timing dynamics really matter here because these massive waves of training are being adopted by early adopters, and that's driving a lot of Astera's business back to them. How inferencing shakes out the dynamics of, "Can you move the H100s and H200s and the inferencing and focus on the B200s and the GB200s for training, and then what happens when NVIDIA's purportedly R200 stuff comes out in 2026, which is the next-generation Blackwell?", this is all public, they're talking about the R series, which might be the next one, and I don't know whether it'll be R or not, there's going to be some sort of lag.

Training will start to come down as inferencing grows. That timing dynamics, I don't think, is worked out or really well understood. With the investment level that's going into this, you would think there'd be tremendous pressure to get inferencing up and running and get these models into production. I'm betting it's shorter. This trough between inferencing and training will be shorter, but a lot goes into that in terms of how fast companies can deploy it and really operationalise it. Regulation could be a threat there, so much may hang on the election and things like that. It's a pretty dynamic equation. Again, the economic pressures will really push the deployment of this technology, and I think companies may start to try to get some of the benefits early. Again, that would have a very beneficial effect on Astera because I think a lot of their future is tied to how this inferencing market develops. Due to the Nvidia GB200 and all, but just the growth of the market as well as their ability to fight off Broadcom will depend on how fast that market grows because the faster it grows, the more, I think, it benefits Astera.

Analyst:
That makes sense because the installed base income you see effect, will just compound baseline.

Specialist:
Exactly, yes. It's a compounding effect. If it's delayed, that gives Broadcom more time to eat away at the market and then, and then they're more tied to the training demand, which they don't have as big advantage. Although, their cable business may ultimately play a role in that training, too.

Analyst:
A few earnings calls ago, Nvidia said that 40% of their GPU demand is already inferencing. How did you interpret that, because most people's base assumption is that it's, by and large, training today. How does one even possibly square the circle on?

Specialist:
A few earnings calls ago, look at their T4 series, the Tesla series, that T4 is venerable, and it's been out there, used in lots of discriminative use cases. Again, inferencing is a lot bigger in computer vision. It's funny to think of us referring to CNNs as traditional or legacy technology. I always laugh when people say that that technology is going to continue to grow in tremendous rates and be incorporated with generative approaches. It wouldn't surprise me if they said that. I would say, right now, looking back at 2023 and 2024, I think 40% is a bit aggressive on their side. They're very focused on thinking about inferencing. There's some training there, distributed training, things like that, but our focus is inferencing. I'm sceptical that 40% of their stuff right now is inferencing. I would be very sceptical of that. Their A100 business and all that that largely focused on computer vision and recommenders and things like that, sure, they probably had a high propensity of inferencing, but you can just look at the growth for generative AI, and that's all training. That is literally all training. When I say all, a high percentage of that is training, very high.

Analyst:
How much of Nvidia's NIMs are going to democratise inferencing use cases? I understand it's probably pretty early in terms of seeing how broad the interest is for customers to use the inferencing microservices that they're now offering, but is that going to accelerate the deployment of these foundation models?

Specialist:
I think it will, the way they're thinking about it. A lot of the stuff they were doing before, they really didn't let you customise it. With the NIMs, they're going to be a lot more flexible on how they roll this out. They're thinking very aggressively here to really embed their software technology into the deployment of these models and the capabilities. With the Riva stuff and NeMo and their frameworks, they've done a really good job there, in my opinion. They're putting it all up in their NGC cloud for availability. Again, I think their strategic focus is they've won the training market, they're going to face a lot more competition in inferencing. There are other big players out there that already have pretty solid inferencing solutions. They think their competitive advantage would be in these software

frameworks, and they're right. Software is very sticky.

Analyst:
Does that mean that AMD has a lot to do in terms of the software ecosystem to actually matter in inferencing, even if they are designing their GPUs towards inferencing over training use cases?

Specialist:
It's like the proprietary networking trends. People don't like lock-in because it drives up cost and causes lots of challenges. There's a big push to make the models, especially in inferencing, the capabilities required for inferencing, to make them more standardised, so they're portable among the ecosystems. Yes, I think they've got work to do in that space, but they're already doing that work. Everyone knows that. The Onyx ecosystem is growing. Yes, I think they've got work to do and they know that. Yes, they're already doing that work.

Analyst:
We've covered the two main concerns for Astera, with largely it being the market slowing down that gives Broadcom more chance to catch up, and maybe the NVL72 will have a little impact on the training side because some people are misunderstanding that those retimers are in the back end, but it's not going to just be 100% of training demand because you still have B100s and B200s doing a lot of that. Maybe it's not as big of an issue as some might fear. Aside from that, they're doing well from a technical standpoint, and CXL is just gravy on top. Is that right?

Specialist:
That's relatively correct. One of the thing, I think, you could really source and support that is just look at the timing of how the GB200 and B200s are going to roll out. If the B200 beats the GB200 to market by 6-9 months, it's going to grab a lot of that demand. Everybody is chasing rack density, don't get me wrong, but then how fast will inferencing grow? Those are the two dynamics that Astera has got to manage. Astera is a solid company. I love to see companies that invest early. They were focused in an area, and it really paid off for them.

Analyst:
What are your thoughts on GB200 having overheating issues as of late? There's a company out of Asia, I was talking about that being a potential concern if they don't fix it by fall. What are your thoughts?

Specialist:
I think we've got early GB200 stuff here, and I'm sure Supermicro does, but there's not a lot of them out yet. We may not even have them yet. I got to check with the guys in the lab that are doing the training side. I'm on the edge side. It'd be pretty amazing that a small company in Asia had their hands on a GB200, first of all. I'd be somewhat sceptical that they've had that. They're all direct liquid-cooled. Direct liquid cooling is required for the GB200. If you're liquid-cooling, you still have forced air for the CPUs and memory and stuff like that. When it's direct liquid-cooled, it's usually the GPUs and maybe the memory sockets, I don't know, memory will generate a lot of heat there. Again, I'd be a bit sceptical. A, it's really early on to say that just where they're at. They're still one year away from coming out in big numbers. Then secondly, with liquid cooling, I don't foresee that that will be a problem.

Analyst:
These are standard ramp-up hiccups.

Specialist:
Yes. It's a 208 billion-transistor chip. I would say, one of the things I've been thinking about in technical risk is, "How about the limitations of on-chip failures and measuring voltage, all that?" I think it was

actually GTC in Jensen's keynote. He mentioned they had technology and optics on that problem, that they were looking at how to manage it at the chip level, both the thermals as well as voltage and current. That's a new area that's definitely going to be required, and there are companies out there that already play in that space.

Transcription ends